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(1) Research Program/Project/Study Title: Blue Swimming Crab (<i>Portunus pelagicus</i>) By-Products as Potential Fish Feed Ingredient: Its Proximate Composition and Partial Micronutrient Profile	
Author's Name:	ROBERT CHRISTIAN F. PATANI
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	REGINA O. DACLES; RINA O. DACLES; ALESSANDRA NICOLE FALALIMPA (Contractual Faculty)
Campus/Satellite College/Address:	Capiz State University - Pontevedra Campus
Telephone/Cell Phone/Fax Number:	09509372319
E-mail Address:	rcfpatani@capsu.edu.ph

Program or Project has a component (if no proceed to 2)?

☐ Yes

☐ No

Component Title:	Evaluation of the proximate composition of selected insects, local agricultural pests, and aquatic invasive species as potential alternatives to fishmeal in the growth of juvenile tilapia, <i>Oreochromis niloticus</i> , and African catfish, <i>Clarias gariepinus</i> .
Component Leader:	GELENE RODA I. ALVAREZ
Academic Rank:	Instructor 1

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFR Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
Municipality/City:		Panay							
Province:		Capiz							
Region:		VI							
(5) Classification <table border="0"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/ Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/ Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/ Commercialization								
(7) Sector/Commodity Crustaceans by-products / Seafood by-products / Animal by-products		(8) Discipline/Field of Specialization Fisheries Science							

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The processing of Blue Swimming Crab (*Portunus pelagicus*) produces a great surplus of by-products, that can be segregated into crab shells and crab viscera. These by-products can be transformed into high-value products and ingredients for various applications, yet these must be characterized. Hence, the present study aimed to determine the complete proximate composition and partial micronutrient profile of blue swimming crab, *Portunus pelagicus* by-products. This study analysed three treatments namely: crab shell (100%), crab viscera (100%), and mixed crab by-product (50:50 w:w). The results of this study showed that the pH levels (9.23 ± 0.09) and ash content ($54.77\% \pm 4.07$) of crab shell by-products are significantly higher compared to the other treatments. Moreover, protein ($46.33\% \pm 1.44$), lipid (3.30 ± 0.21), and Magnesium (513.67 ± 0.67) were significantly higher among all sample treatments. The results indicate that the crab viscera by-product of blue swimming crabs (*Portunus pelagicus*) holds nutritional value suitable for feed formulation. Crab viscera by-product, being rich in protein and fat, can serve as a valuable ingredient in animal feed formulations, particularly for aquaculture.

(9B) Keywords: (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include the process, or methodology employed.)

Portunus pelagicus, Crab Shell By-Product, Crab Viscera By-Product, Proximate Composition, Micronutrient Profile

(10) Introduction (This includes the rationale, background of the study, and objectives, which are not divided by subheadings. Incorporate at least three (3) reviews).

According to FAO (2022), 89% or 157 million tonnes of fisheries and aquaculture production of aquatic animals were utilized for direct human consumption, while roughly 9% or 16 million tonnes were converted into fishmeal and fish oil. Aquaculture in the Philippines, driven by the need for food security, relies heavily on fishmeal as a primary animal-based protein source in aquafeed. However, challenges such as limited availability and high costs brought about by transporting and processing ingredients have prompted the exploration of cheaper alternatives. In recent years, the utilization of fish or seafood by-products as an alternative source of protein and lipids for the aquafeed industry has also been a practice (Bechtel et al., 2007; Olsen et al., 2014).

Seafood by-products mainly refer to material parts that are left over following the preparation of the main product (Stevens et al., 2017). This may include trimmings, skins, heads, frames (bones with attached flesh), viscera, and blood. These by-products may constitute roughly 70% of fish and shellfish after processing, and these were mainly converted into value-added commercial products (Olsen et al., 2014). Specifically, based on the area, species, and processing techniques, crustacean by-products may be composed of heads, shells, gills, pleopods, abdominal flaps, tails, eye stalks, and viscera. These residual materials are found to be excellent sources of active ingredients such as protein (25-50%), chitin (25-35%), minerals (15-35%), lipids (0.2-17%), and pigments (Hamed et al., 2015; De Aguiar Saldanha Pinheiro et al., 2021). The valorization of these materials makes them an available resource in the concept of circular economy, where instead of being disposed of or incinerated, they are transformed into high-value products, hence reducing waste and maximizing resource utilization (Zhang et al., 2024).

Crustaceans form a major fishery with good market potential in Asian countries. Undoubtedly, a staggering amount of 6 million to 8 million tonnes of crustacean (lobster, shrimp, and crab) waste is produced, whereas 1.5 million tonnes of this waste is mainly recovered in Southeast Asia alone (Yan & Chen, 2015). Blue swimming crab (BSC), *Portunus pelagicus*, locally known in the dialects as *kasag*, *alimasag*, *lambay*, or *masag*, is a Portunid crab widely distributed throughout the Indo-Pacific region and became a significant fishery commodity because it has become an essential economic contributor in countries within the region, such as the Philippines due to its global trade (Romano and Zeng, 2008). Region VI, which faces the Visayan Sea, is the country's largest contributor to BSC production, mainly accounting for the high production from the municipal fisheries sector (PSA, 2022). Locally, Capiz accounted for a total of 1,181.06 MT of BSC production for the year 2021.

Since BSC is a highly valued species in the Philippines, it is also widely processed in various regions. The processing of blue swimming crabs plays a significant role in the seafood industry, contributing to

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the local economy and the export market. In Panay, Capiz, there is a thriving small-scale crab processing plant facility that processes *Portunus pelagicus* by removing crab meats from the exoskeleton. This processing produces a significant surplus of by-products, where the facility further segregates the crab shells and crab viscera. Crab shells are usually dried and stored and/or sold to interested buyers at an affordable price, while crab viscera are either dried and dumped or fed to poultry or swine (Ken-RJ, pers. comm.). With the indigenous application of the by-products produced by the local processing facility, it is essential that the BSC by-products, specifically the crab shells and crab viscera by-products, be characterized with their proximate composition and micronutrient profile further to understand their possible application in the fish feed industry.

Hence, this study aimed to determine the proximate composition (Moisture, Crude Protein, Crude Lipids, Crude Fiber, NFE, & Ash Content) and partial micronutrient profile (Magnesium and Calcium) of blue swimming crab, *Portunus pelagicus*, by-products (100% Crab Shell; 100% Crab Viscera; 50:50% Mixed Crab Shell and Viscera). Moreover, this study also aimed to determine if there is a significant difference in the proximate composition and partial micronutrient profile of blue swimming crab, *Portunus pelagicus* by-products.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science research and methodology in Social Sciences. The presentation of materials may not necessarily be in separate headings but may be incorporated into the specific methods used. Justify paragraphs with main headings in caps.)

SAMPLE COLLECTION & PREPARATION

A total of 38 kgs of crab by-products (12 kgs of crab shell and 26 kgs of crab viscera by-products) were collected from KEN-RJ Marine Products from Agojo, Panay, Capiz (11°36'28.82"N and 122°48'36.77"E) from April to May 2024. Crab shell by-products include all the crab exoskeleton after processing, such as the carapace, thorax, and appendages. In contrast, crab viscera by-products include the digestive system (hindgut in the abdomen flap), mouth parts, reproductive organs, abdomen flap, and gills. Samples were brought to the Fisheries Department, College of Agriculture and Fisheries, CAPSU-Pontevedra, Bailan, Pontevedra 5802, Capiz. Each sample was washed with tap water to remove debris, sundried for 24 hours to remove moisture, grounded using a mixer grinder to achieve a powdered sample, stored in a plastic bag at room temperature, and subjected for analysis. The pH level of the samples was described and identified using the electrometric method.

The following treatments were prepared that were subjected to analyses: 100% Crab Shell By-Products (CS), 100% Crab Viscera By-Products (CV), and mixed equal parts of crab by-products (M). All the analyses were conducted in triplicates.

PROXIMATE COMPOSITION ANALYSIS

The moisture content, % ash, crude protein, crude lipids, crude fiber, and NFE of the different treatments of crab by-products. Moisture content was analyzed at the CAPSU Soils Laboratory, CAPSU-Pontevedra. Moisture content is expressed as the amount of water as a percentage (%); the remaining portion is the dry matter content. Ten (10) grams of each sample were analyzed using a moisture analyzer for 50 minutes at 150°C. Moreover, the crude protein, crude lipid, and ash composition were analyzed at the Nutrition Laboratory, Multispecies Hatchery, Institute of Aquaculture, College of Fisheries and Ocean Science, UP Visayas, Maiag-ao 5023, Iloilo, Philippines, while crude fiber was analyzed at the Institutional Research and Development Office, Saint Gabriel College, Kalibo 5600, Aklan. The protein content was determined using the Macro-Kjeldahl Method, while the ash was determined by burning off the organic matter at 550°C in a muffle furnace (AOAC, 2016). Meanwhile, the simple adaption and economical method of Bligh and Dyer (AOAC, 2016) was carried out to determine the lipid content of the samples. Crude fiber was analyzed using the gravimetric method (AOAC, 2016). Lastly, Nitrogen-free extracts (NFE) content was computed by difference, where all other components were subtracted from 100.

MICRONUTRIENT PROFILE ANALYSIS

A partial micronutrient profile, specifically for Calcium (Ca) and Magnesium (Mg), was analyzed for the different samples of crab by-products. The analysis for Ca and Mg followed the CAPSU Soils Laboratory protocol, where the sample solution's absorbance was measured using a spectrophotometer.

DATA ANALYSIS PROCEDURE

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All data collected were encoded and organized in a Microsoft Excel Spreadsheet. Using the free JAMovi software, the assumptions for the Normality and Homogeneity of each analysis result were tested using Shapiro-Wilk's and Levene's tests, respectively. Once assumptions were met, a one-way Analysis of Variance (ANOVA) was used to test the significant difference among the treatments for each analysis parameter. Tukey's Test was used for post-hoc analysis of parameters that showed significant differences. However, in analysis where assumptions for normality and homogeneity were not satisfied, the non-parametric counterpart for one-way ANOVA (Kruskal-Wallis 1-way ANOVA) was used. A pairwise comparison was then used to determine the significant difference between treatment means. All statistical analyses were conducted at a 95% level of significance.

(12) Results and Discussions (This part contains a presentation of results (data) with reference to the table and figure, followed by a discussion with support from the reviewed literature. The sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions, below table headings, and above major subdivisions. Justify paragraphs with main headings in caps.)

PROXIMATE COMPOSITION OF CRAB BY-PRODUCTS

The proximate composition of blue swimming crab by-products (mean $\pm$ SEM) is presented in Table 3. Results showed that all samples had basic pH levels. The analysis of variance revealed that there is a significant difference ($p < 0.05$) in the pH levels among the samples. Post Hoc analysis further confirmed that the pH level of crab shell by-product powder differed significantly. At the same time, there was no significant difference between the pH levels of crab viscera and mixed crab by-product treatments.

Moreover, the highest moisture content was characterized from the crab viscera by-product ($8.58\% \pm 0.12$), and moisture content was lowest at the crab shell by-product (6.53 ± 0.60). However, analysis of variance showed that there is no significant difference in the moisture content of all the treatments. The highest protein and lipid content were observed in the crab viscera by-products with a percent value of 46.33 ± 1.44 and $3.30\% \pm 0.21$, respectively, while the lowest ash content was observed with the same sample treatment. Analysis of variance revealed that the protein, lipid, and ash content of crab viscera by-products were significantly different among all the treatments. While crude fiber was highest in the mixed sample treatment, analysis of variance revealed that there is no significant difference in the crude fiber content of the BSC by-products. Lastly, NFE was highest in the crab shell by-products (20.13%), while it was lowest in the crab-viscera by-products (12.73%).

Table 1. Complete proximate composition of Blue Swimming Crab (*Portunus pelagicus*) by-products.

Parameter	Crab Shell	Crab Viscera	Mixed
pH Level	9.23 $\pm$ 0.09 ^a	8.50 $\pm$ 0.06 ^b	8.61 $\pm$ 0.13 ^b
% Moisture	6.53 $\pm$ 0.60 ^a	8.58 $\pm$ 0.12 ^a	7.08 $\pm$ 0.07 ^a
Crude Protein	23.77 $\pm$ 1.37 ^a	46.33 $\pm$ 1.44 ^b	33.98 $\pm$ 0.30 ^c
Crude Lipid	0.98 $\pm$ 0.18 ^a	3.30 $\pm$ 0.21 ^b	1.41 $\pm$ 0.07 ^a
Crude Fiber	9.91 $\pm$ 0.54 ^a	10.26 $\pm$ 0.92 ^a	11.30 $\pm$ 0.29 ^a
NFE	20.13	12.73	16.85
Ash Content	54.77 $\pm$ 4.07 ^a	30.11 $\pm$ 1.22 ^b	40.99 $\pm$ 1.99 ^a

Note: Values are presented as mean $\pm$ SEM, except for NFE. Superscripts of the same letter per row (parameter) are not significantly different ($p > 0.05$).

Utilizing fish/seafood by-products to produce fishmeal, protein hydrolysates, and fish oil indirectly contributes to healthy food production, with the rapidly growing aquaculture industry being the primary consumer of these products. In relation to this, the present study on the characterization of Blue Swimming Crab (*Portunus pelagicus*) by-products highlighted key findings. Comparatively higher levels of protein and lipids ($46.33\% \pm 1.44$ and $3.30\% \pm 0.21$, respectively) were observed in crab viscera by-products in contrast to crab shell by-products or in the combination of both.

The protein content of the crab by-products in this study is comparable to the 25-50% protein content of crustacean by-products, as discovered in the analysis of De Aguiar Saldanha Pinheiro et al. (2021). The present values for protein content of BSC by-products are higher than the protein content of the by-products other species such as *Chionoecetes opilio* shell and discard with 33-35% (Vilasoja-Martinez et al., 2007), *Birgus latro* shell powder with 16-21% (Rasulu et al., 2021), *Portunus segnis* by-product with 2.52% (Jabuer et al., 2021), *Cardisoma armatum* shell with 13% (Elegbede & Fashina-Bombata, 2013), and *Callinectes pallidus* shell with 13-16% (Elegbede & Fashina-Bombata,

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2013; Ibrahim et al., 2018). Even the lowest protein content recorded for this crab shell ($23.77\% \pm 1.37$) of this study is comparably higher than the protein content characterized in the crab shell powder of BSC (8.17%) in the works of Amelia et al., 2023, the closest related study for this work. Furthermore, the protein content of the crab by-products found in this study is also comparable to the characterized protein content of the purely edible crab meat portion of various crab species, such as *Scylla serrata* claw and body meat with 12.35% (Siddique et al., 1987), *Portunus sanguinolentus* claw and body meat with 13.09% (Siddique et al., 1987), *Callinectes sapidus* meat with 14-15% (Gokodlu & Yerlikaya, 2003), *Carcinus maenus* meat with 16-18% (Skonberg & Perkins, 2002), and *Cancer pagarus* muscle with 16-21% (Barrento et al., 2010).

In terms of the lipid contents, the lipidic content in crustacean by-products may range from 0.2-17% (De Aguiar Saldanha Pinheiro et al., 2021). More specifically, the results found in the present study are comparable to the lipid content of the by-products of other species as found in the works of Rasulu et al., 2021 (1.28-2.02%), Jabuer et al., 2021 (0.57%), Amelia et al., 2023 (0.26-0.28%), and edible crab meat portions as found in the work of Siddique et al., 1987 (1.02-1.29%), Skonberg & Perkins, 2002 (0.5-1.2%), Gokodlu & Yerlikaya, 2003 (0.64-1.21%), Barrento et al., 2010 (0.2-0.4%), and Pathak et al., 2021 (2.17-3.71%). However, these values are relatively lower compared to the characterized lipid content of the various crab species such as *Callinectes pallidus* shell with 6.75-24% (Elegbede & Fashina-Bombata, 2013, Ibrahim et al., 2018), *Cardisoma armatum* shell with 24% (Elegbede & Fashina-Bombata, 2013), and *Chionoecetes opilio* by-products with 14.82% (Beaulieu et al., 2009).

The seafood processing industry generates substantial volumes of byproducts and discards that encompass a range of materials from marine organisms, serving as sources of lipids, proteins, pigments, small molecules, and chitinous materials. While these exist as waste residues, they carry significant nutritional and functional value. Researchers have been exploring water-soluble derivatives of chitin due to the water insolubility challenge on a large scale (Du J., Tan E., Kim H.J., Zhang A., Bhattacharya R., Yarema K.J., 2013).

In this study, crab viscera by-products is found to be a promising source of active ingredients with suitable amounts ideal for feed formulation in the aquaculture industry. This can be attributed to the fact that crab viscera by-products are composed of the hepatopancreas and digestive tract organs which play crucial roles in digestion, nutrient absorption, and metabolism. Various chemical analyses have revealed the composition of crab meals and crab protein concentrates, highlighting the protein, fat, ash, and chitin content in these products. This underscores their nutritional significance and potential applications in food and feed industries (Husby, 1980; Johnson, 1988; Nicholson et al., 1996).

Waste characterization analysis conducted by the researchers unveiled the composition of waste materials generated during processing, ranging from crab shells to visceral waste, residual meat, soft tissues, and other byproducts. The utilization of marine invertebrate byproducts as food and feed supplements on a global scale presents opportunities to enhance diets, reduce waste, and improve profitability, offering health-promoting ingredients like marine oils, calcium, enzymes, antioxidants, flavorings, and pigments (Obatolu, Skonberg, Camire, & Dougherty, 2005; Tibbetts, Olsen, & Lall, 2011; Velu & Munuswa-my, 2007). These byproducts have demonstrated diverse applications in the food industry and beyond, contributing to various sectors including health, nutrition, aquaculture, and cosmetology (Beaulieu, Thibodeau, Bryl, & Carbonneau, 2009).

The increasing demand for products derived from fishing has fueled aquaculture activities and the necessity for suitable fish feeds. Incorporation of crustacean-processing waste and byproducts in animal feeds has been a common practice due to their nutrient density and capacity to enrich diets and create value-added products (Heu, Kim, & Shahidi, 2003; Mach, Nguyen, & Nortvedt, 2010; Murphy et al., 2003; Oliveira Cavaleiro, Oliveira de Souza, & Bora, 2007; Rosa & Nunes, 2003; Shahidi & Synowiecki, 1991). These byproducts offer a diverse array of health-promoting components and minerals, playing a significant role in feed formulations and value addition across industries.

PARTIAL MICRONUTRIENT PROFILE OF CRAB BY-PRODUCTS

Table 2 presents the partial micronutrient profile (mean $\pm$ SEM) of Blue Swimming Crab (*Portunus pelagicus*) waste samples. Results showed that there is a significant difference in the Magnesium (Mg) levels of the sample treatments, where it was significantly higher in the crab viscera by-products (513.67 ± 0.67). Calcium (Ca) level was also highest in the crab viscera by-products $2,651.00 \pm 108.49$, however analysis of variance revealed that there is no significant difference in the Ca of the BSC by-products

Table 2. Partial Micronutrient Profile of Blue Swimming Crab (*Portunus pelagicus*) by-products.

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Sample Treatment	Mineral Content from Ash (in PPM)	
	Magnesium (Mg)	Calcium (Ca)
Crab Shell	501.33 ± 0.33 ^a	2,172.00 ± 11.50 ^a
Crab Viscera	513.67 ± 0.67 ^b	2,651.00 ± 108.49 ^a
Mixed	506.67 ± 0.33 ^c	2,121.33 ± 180.14 ^a

Note: Values are presented as mean ± SEM. Superscripts of the same letter per column are not significantly different ($p > 0.05$).

The findings of the present study challenge the widely held belief that crab shells are rich sources of calcium due to their main constituent, calcium carbonate (CaCO₃). Previous research has consistently indicated that the highest calcium content in crabs is found in their shells (Kathirvel et al., 2014; Haryati et al., 2019). Similarly, Ahmed (2022) reported elevated levels of magnesium in crab shells compared to crab meat. The disparity in results between this study and previous findings may be attributed to human errors and the appropriateness of the methodology used. Calcium and magnesium are essential elements for fish function, and the substantial presence of these elements in crab waste (specifically, the carapace and viscera) suggests promising prospects for utilizing crab waste in the food and feed industry. Overall, the composition of active ingredients within crab by-products vary depending on the species, the location where the species were collected, the time of collection, and the technique of processing (Olsen et al., 2014), however, the results of the present study which focused on the characterization of the by-products of blue swimming crab is promising.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions.

The study reveals that blue swimming crab by-products from processing waste can be used in feed formulation to create high-quality feeds for various aquatic species. These by-products, including shells and heads, provide essential nutrients and contribute to animal growth and development. The crab viscera, high in protein and fat, is suitable for feed formulation, especially in aquaculture and livestock sectors. By incorporating crab viscera into feed compositions, manufacturers and farmers can reduce feed costs, efficiently utilize by-products, and create sustainable, nutritionally enriched feeds for livestock and aquaculture species. Regulatory compliance, market research, sustainable methods, resource efficiency, waste reduction, and environmental stewardship are promoted.

(14) Output (Indicate the specific products, processes, or services which the project has produced; quantify when possible.)

Information on the proximate composition and micronutrient profile of blue swimming crab by-products (crab shell by-product and crab viscera by-product powder).

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities).

N/A

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Nature of Expenses	Quantity	Amount/Unit	Amount
A. Personel Expenses	n/a	n/a	n/a
B. Maintenance and Other Operating Expenses (MOOE)			
- Laboratory/Experiment Supplies		1,000.00	1,000.00
- Professional Fees (Laboratory Fee for Proximate Analysis)	3 samples / 3 replicates	5,020.00	15,060.00
- Professional Fees (Laboratory Fee for Micronutrient Profiling)	2 analysis for 3 samples	400.00	400.00
- Operating cost for grinding			500.00
C. Capital Outlay	n/a	n/a	n/a
Total			16,960.00

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(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

1. Prof. Gelene Roda I. Alvarez
2. CAPSU-Pontevedra Research Office
3. Institute of Aquaculture, College of Fisheries and Ocean Sciences, UP Visayas, Miag-ao
4. Saint Gabriel College Inc., Kalibo, Aklan
5. CAPSU Soils Laboratory, Capiz State University - Pontevedra

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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